Methods

Preview

1. Data: acoustic/articulatory data from published datasets
2. Distinctiveness measurement in acoustics & articulation
3. Bigram surprisal calculation
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target

Methods

Data

- Acoustic and production data in British English, American English, and French, collected with ultrasound/optical imaging, EMA, and MRI, were used:

	Corpus	Language	Instrument	# participants	# data points
BE-US	Tongue and Lips Corpus (Ribeiro *et al*. 2021)	British English	Ultrasound/ Optical lip imaging	41	251,722
FR-MRI	ArtSpeechMRIfr (Douros *et al*. 2019)	French	MRI	10	42,644
AE-EMA	USC-TIMIT (Narayanan *et al*. 2014) & 75-Speaker Annot-16 (Shi *et al*. 2025)	American English	EMA	4 (subset)	81,698
AE-MRI			MRI	17	232,088
AE-MRI-annotated				7 (subset)	42,218
			Total:	68	650,370

Ribeiro, M. S., Sanger, J., Zhang, J.-X., Eshky, A., Wrench, A., Richmond, K., & Renals, S. (2021) TaL: A synchronised multi-speaker corpus of ultrasound tongue imaging, audio, and lip videos. In *Proceedings of the ieee workshop on spoken language technology*. Shenzhen, China. Narayanan, S., Toutios, A., Ramanarayanan, V., Lammert, A., Kim, J., Lee, S., Nayak, K., Kim, Y.-C., Zhu, Y., Goldstein, L., Byrd, D., Bresch, E., Ghosh, P., Katsamanis, A., & Proctor, M. (2014). Real-time magnetic resonance imaging and electromagnetic articulography database for speech production research (TC). *The Journal of the Acoustical Society of America, 136*(3), 1307–1311. Douros, I.K., Felblinger, J., Frahm, J., Isaieva, K., Joseph, A.A., Laprie, Y., Odille, F., Tsukanova, A., Voit, D., Vuissoz, P.-A. (2019) A Multimodal Real-Time MRI Articulatory Corpus of French for Speech Research. *Proc. Interspeech 2019*, 1556-1560. Shi, X., Zhang, Y., Lu, Y., Ma, M., Feng, T., Toutios, A., Hsu, H., Goldstein, L., Narayanan, S. (2025) 75-Speaker Annot-16: A benchmark dataset for speech articulatory rt-MRI annotation with articulator contours and phonetic alignment. *Proc. Interspeech 2025*, 2175-2179.